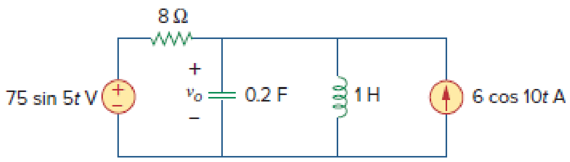
Chapter 10, Problem 6PP

Problem

Calculate *vo* in the circuit of Fig. 10.15 using the superposition theorem.

Figure 10.15:



Step-by-step solution

Step 1 of 11

Consider the following circuit diagram:

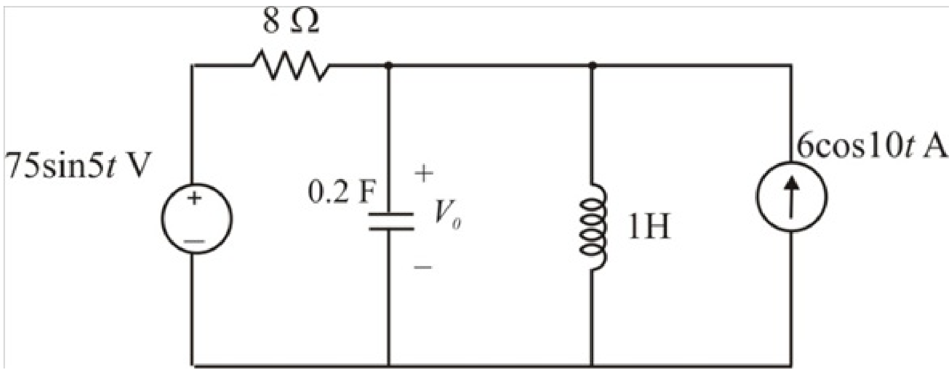


Figure 1

Step 2 of 11

Let us assume that $\mathbf{V}_o = \mathbf{V}_o' + \mathbf{V}_o''$.

According to superposition theorem $\mathbf{V}_o'$ and $\mathbf{V}_o''$ are the voltages due to voltage and current sources respectively.

Consider the following circuit in which current source is kept open circuit to find $\mathbf{V}_o'$.

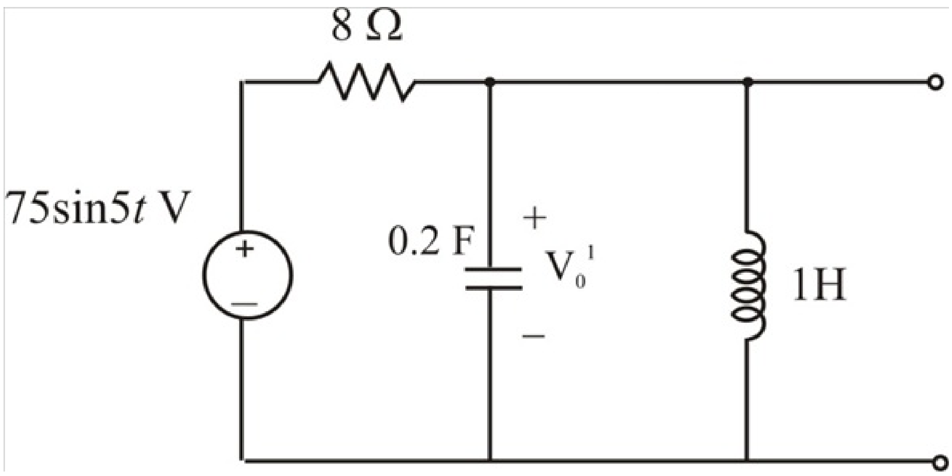


Figure 2

Step 3 of 11

Convert the circuit to frequency domain.

$$75\sin 5t = 75\angle -90°$$

And

$$\omega = 5 \text{ rad/s}$$

Convert the inductance value.

$$\begin{aligned} L &= 1 \text{ H} \\ j\omega L &= j(5 \text{ rad/s})(1 \text{ H}) \\ &= j5\ \Omega \end{aligned}$$

Convert the capacitance value.

$$\begin{aligned} C &= 0.2 \text{ F} \\ \frac{1}{j\omega C} &= \frac{1}{j(5 \text{ rad/s})(0.2 \text{ F})} \\ &= \frac{1}{j1}\ \Omega \\ &= -j1\ \Omega \end{aligned}$$

Step 4 of 11

Draw the equivalent frequency domain circuit:

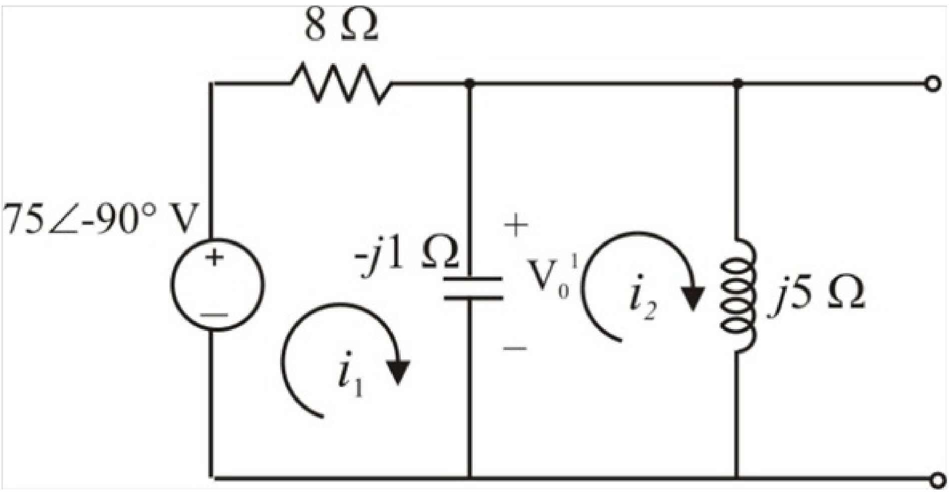


Figure 3

Step 5 of 11

Assume that the currents $\mathbf{I}_1$ and $\mathbf{I}_2$ are flowing in the circuit.

Apply Kirchhoff's voltage law to mesh 1.

$$8\mathbf{I}_1 - j(\mathbf{I}_1 - \mathbf{I}_2) = 75\angle -90^\circ$$

$$(8-j)\mathbf{I}_1 + j\mathbf{I}_2 = -j75 \text{ ...... (1)}$$

Apply Kirchhoff's voltage law to mesh 2.

$$-j(\mathbf{I}_2 - \mathbf{I}_1) + j5\mathbf{I}_2 = 0$$

$$j\mathbf{I}_1 + j4\mathbf{I}_2 = 0 \text{ ...... (2)}$$

Write equations (1) and (2) in the matrix form.

$$\begin{bmatrix} 8-j & j \\ j & j4 \end{bmatrix}\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} -j75 \\ 0 \end{bmatrix}$$

Calculate the determinant values of matrices to find $\mathbf{I}_1$ and $\mathbf{I}_2$.

$$\begin{aligned} \Delta &= \begin{vmatrix} 8-j & j \\ j & j4 \end{vmatrix} \\ &= (8-j)(j4) - (j)(j) \\ &= j32 + 4 + 1 \\ &= 5 + j32 \end{aligned}$$

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} -j75 & j \\ 0 & j4 \end{vmatrix} \\ &= (-j75)(j4) - (j)(0) \\ &= 300 \end{aligned}$$

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} 8-j & -j75 \\ j & 0 \end{vmatrix} \\ &= (8-j)(0) - (-j75)(j) \\ &= -75 \end{aligned}$$

Step 6 of 11

Calculate the current $\mathbf{I}_1$ in mesh 1:

$$\begin{aligned} \mathbf{I}_1 &= \frac{\Delta_1}{\Delta} \\ &= \frac{300}{5+j32} \\ &= 9.26\angle -81.11^\circ \text{ A} \end{aligned}$$

Calculate the current $\mathbf{I}_2$ in mesh 2:

$$\begin{aligned} \mathbf{I}_2 &= \frac{\Delta_2}{\Delta} \\ &= \frac{-75}{5+j32} \\ &= 2.31\angle 98.88^\circ \end{aligned}$$

Calculate the value of voltage $\mathbf{V}_o'$:

$$\begin{aligned} \mathbf{V}_o' &= (\mathbf{I}_1 - \mathbf{I}_2)(-j1\,\Omega) \\ &= (9.26\angle -81.11^\circ - 2.31\angle 98.88^\circ)(1\angle -90^\circ) \\ &= 11.57\angle -171.11^\circ \end{aligned}$$

Convert the voltage value $\mathbf{V}_o'$ to time domain.

$$\begin{aligned} \mathbf{V}_o' &= 11.57\cos(5t - 171.11^\circ) \\ &= 11.57\sin(5t - 81.11^\circ) \end{aligned}$$

The value of voltage v_o' is $11.57\sin(5t - 81.11^\circ)$ V.

Consider the following circuit in which voltage source is kept short circuit to find $\mathbf{V}_o''$.

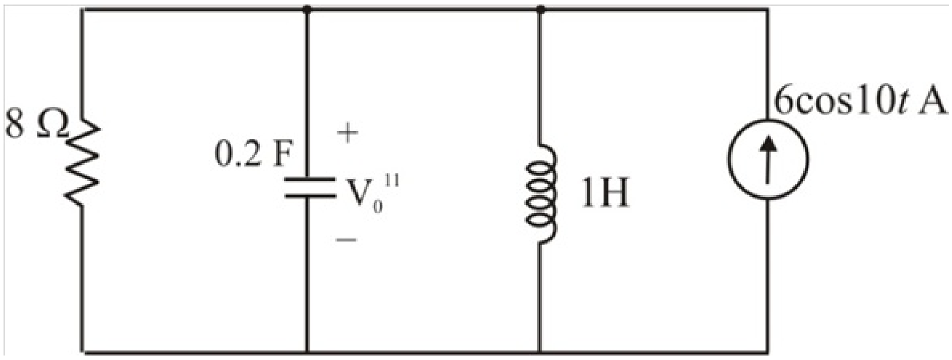


Figure 4

Convert the circuit to frequency domain.

$$6\cos 10t = 6\angle 0°$$

And

$$\omega = 10 \text{ rad/s}$$

Convert the inductance value.

$$\begin{aligned} L &= 1 \text{ H} \\ j\omega L &= j(10 \text{ rad/s})(1 \text{ H}) \\ &= j10\ \Omega \end{aligned}$$

Convert the capacitance value.

$$\begin{aligned} C &= 0.2 \text{ F} \\ \frac{1}{j\omega C} &= \frac{1}{j(10 \text{ rad/s})(0.2 \text{ F})} \\ &= \frac{1}{j2}\ \Omega \\ &= -j0.5\ \Omega \end{aligned}$$

Draw the equivalent frequency domain circuit:

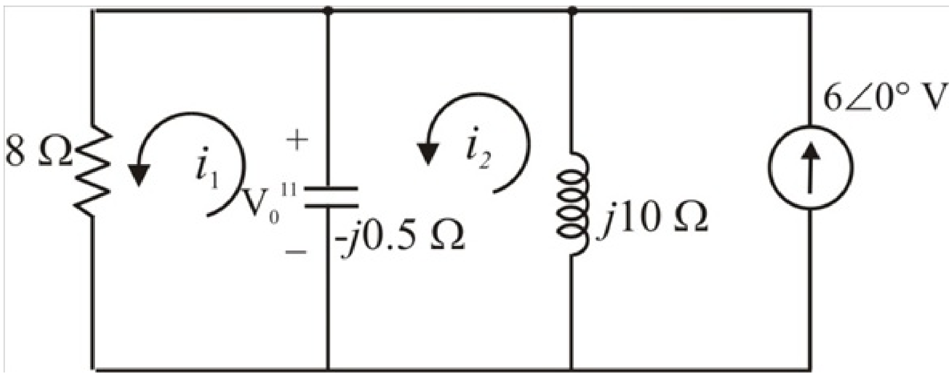


Figure 5

Assume that the currents $\mathbf{I}_1$ and $\mathbf{I}_2$ are flowing in the circuit.

Apply Kirchhoff's voltage law to mesh 2.

$$(\mathbf{I}_2-6)(j10)-j0.5(\mathbf{I}_2-\mathbf{I}_1)=0$$
$$(j10-j0.5)\mathbf{I}_2+j0.5\mathbf{I}_1=j60$$

$$j9.5\mathbf{I}_2+j0.5\mathbf{I}_1=j60 \;\; \ldots\ldots \text{ (1)}$$

Apply Kirchhoff's voltage law to mesh 1.

$$8\mathbf{I}_1-j0.5(\mathbf{I}_1-\mathbf{I}_2)=0$$

$$(8-j0.5)\mathbf{I}_1+j0.5\mathbf{I}_2=0 \;\; \ldots\ldots \text{ (2)}$$

Represent the equations (1) and (2) in matrix form.

$$\begin{bmatrix} j0.5 & j9.5 \\ 8-j0.5 & j0.5 \end{bmatrix}\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix}=\begin{bmatrix} j60 \\ 0 \end{bmatrix}$$

Calculate the determinant values of matrices to find $\mathbf{I}_1$ and $\mathbf{I}_2$.

$$\begin{aligned}\Delta &= \begin{vmatrix} j0.5 & j9.5 \\ 8-j0.5 & j0.5 \end{vmatrix} \\ &= (j0.5)(j0.5)-(j9.5)(8-j0.5) \\ &= -5-j76\end{aligned}$$

$$\begin{aligned}\Delta_1 &= \begin{vmatrix} j60 & j9.5 \\ 0 & j0.5 \end{vmatrix} \\ &= (j60)(j0.5)-(j9.5)(0) \\ &= -30\end{aligned}$$

$$\begin{aligned}\Delta_2 &= \begin{vmatrix} j0.5 & j60 \\ 8-j0.5 & 0 \end{vmatrix} \\ &= (j0.5)(0)-(j60)(8-j0.5) \\ &= -30-j480\end{aligned}$$

Calculate the current $\mathbf{I}_1$ in mesh 1:

$$\begin{aligned}\mathbf{I}_1 &= \frac{\Delta_1}{\Delta} \\ &= \frac{-30}{-5-j76} \\ &= 0.393\angle -86.23^\circ \text{ A}\end{aligned}$$

Calculate the current $\mathbf{I}_2$ in mesh 2:

$$\begin{aligned}\mathbf{I}_2 &= \frac{\Delta_2}{\Delta} \\ &= \frac{-30-j480}{-5-j76} \\ &= 6.31\angle 0.187^\circ \text{ A}\end{aligned}$$

Calculate the value of voltage $\mathbf{V}_o''$:

$$\begin{aligned}\mathbf{V}_o'' &= (\mathbf{I}_1 - \mathbf{I}_2)(-j0.5\ \Omega) \\ &= (0.393\angle -86.23^\circ \text{ A} - 6.31\angle 0.187^\circ \text{ A})(0.5\angle -90^\circ) \\ &= (6.302\angle -176.24^\circ)(0.5\angle -90^\circ) \\ &= 3.15\angle 93.75^\circ\end{aligned}$$

Convert the voltage value $\mathbf{V}_o''$ to time domain.

$$\begin{aligned}\mathbf{V}_o'' &= 3.15\cos(10t - 180^\circ + 93.75^\circ) \\ &= 3.15\cos(10t - 86.25^\circ)\end{aligned}$$

The value of voltage v_o'' is $3.15\cos(10t - 86.25^\circ)$ V.

Step 11 of 11

Calculate the value of required voltage v_o:

$$\begin{aligned}\mathbf{V}_o &= \mathbf{V}_o' + \mathbf{V}_o'' \\ &= 11.57\sin(5t - 81.11^\circ) \text{ V} + 3.15\cos(10t - 86.25^\circ) \text{ V}\end{aligned}$$

Hence, the following is the required voltage value v_o.

$$\boxed{(11.57\sin(5t - 81.11^\circ) + 3.15\cos(10t - 86.25^\circ)) \text{ V}}.$$